

Extended Abstract

Motivation Recent advancements in aligning large language models (LLMs) with human preferences have predominantly relied on either reinforcement learning from human feedback (RLHF) or methods like Direct Preference Optimization (DPO) that implicitly infer rewards without explicit modeling. While DPO reduces complexity by eliminating explicit reward models, the availability of high-quality, pretrained reward models offers an opportunity to directly leverage explicit reward signals to potentially achieve better alignment. We explore the potential of directly distilling pretrained reward models into language models through a novel and flexible optimization approach termed Direct Reward Optimization (DRO).

Method DRO builds upon the idea of matching policy ratios to an optimal reference policy defined by a pretrained reward model under a mean-squared error loss. Unlike traditional approaches such as DPO, which implicitly estimate rewards from preference data, DRO explicitly regresses the differences in log-likelihood ratios onto reward differences, and is able to take into account the richer reward signal from a separate reward model. We introduce DRO through a ratio-matching framework, demonstrating it as a generalization of recent reward distillation methods like REBEL. Specifically, DRO generalizes the regression step by allowing arbitrary monotonic activation functions, beyond the logarithmic transform typically used in prior work, offering fine-grained control over training dynamics. Furthermore, DRO provides a unified theoretical perspective that connects the training objectives of DPO, IPO, and REBEL, highlighting fundamental relationships among these methods.

Implementation We implemented SFT, DPO, and DRO from scratch using PyTorch. Our framework supports dynamic adjustment of key hyperparameters, such as the regularization strength, and accommodates exploration of various monotone activation functions. We evaluated DRO against SFT and DPO, using the HuggingFace datasets UltraFeedback-Binarized and Anthropic’s HH-RLHF on the Qwen2.5 0.5B and Pythia 2.8B base models respectively.

Results On the UltraFeedback-Binarized dataset, DRO demonstrates superior performance over established baselines. When fine-tuning the Qwen2.5 0.5B model, DRO achieved a 0.63 win-rate against the Qwen2.5 0.5B-Instruct model and a reward margin of 0.76, outperforming DPO’s 0.54 win-rate and 0.64 reward margin. Pairwise win-rate comparisons confirm DRO as the Condorcet winner against SFT, DPO, and the baseline instruction-tuned Qwen2.5 model. Further validation on the Pythia 2.8B model with the Anthropic HH-RLHF dataset showed DRO’s performance to be highly competitive with DPO, underscoring the generalizability of our approach across different models and datasets.

Discussion The DRO framework provides a unified theoretical lens for reward-model distillation. By deriving the REBEL objective from a general ratio-matching principle, we clarify its connection to other preference alignment methods. Specifically, DRO recasts the REBEL objective as an IPO-like regression, where the target is not a fixed margin but the dynamic, continuous reward difference from a pretrained model. While our framework theoretically supports arbitrary activation functions, our experiments revealed that non-logarithmic functions (e.g., power functions) were numerically unstable due to the vast dynamic range of language model likelihoods. This underscores the practical significance of the logarithm’s compressive properties for the stability and success of this alignment method.

Conclusion Direct Reward Optimization is a robust and effective framework for distilling reward models, delivering superior empirical performance while providing theoretical clarity that unifies REBEL with DPO and IPO. Its success highlights the value of leveraging explicit reward signals in a simple, regression-based objective. Future research should investigate DRO’s sensitivity to different reward models, its integration into online RLHF settings, and the development of numerically stable non-logarithmic transforms to unlock the full potential of the ratio-matching framework.

Direct Reward Optimization for Reward-Model Distillation

Kenan Hasanaliyev

Department of Computer Science
Stanford University
kenanhas@stanford.edu

Schwinn Saereesitthipitak

Department of Electrical Engineering
Stanford University
schwinn@stanford.edu

Rohan Sanda

Department of Electrical Engineering
Stanford University
rsanda@stanford.edu

Abstract

Recent preference alignment methods like Direct Preference Optimization (DPO) have moved away from explicit reward modeling. However, distilling pre-existing, high-quality reward models remains a powerful and efficient alignment strategy. This work introduces **Direct Reward Optimization (DRO)**, a general ratio-matching framework for reward-model distillation. We show that DRO provides a novel, alternative derivation for the recently proposed REBEL algorithm, recovering its objective by selecting the logarithm as the transformation function. This framework unifies REBEL with other alignment methods by formally connecting its objective to the structure of Identity-Preference Reward Optimization (IPO), where the optimization target becomes a dynamic reward difference rather than a fixed margin. We implemented DRO and fine-tuned a Qwen2.5 0.5B model on the UltraFeedback-Binarized dataset. Our experiments demonstrate that DRO outperforms DPO in both pairwise win-rate and reward margin. The DRO framework also provides a principled way to explore alternative transformation functions, opening new possibilities for reward distillation though we were unable to determine good transformation function choices in this project.

1 Introduction

Direct Preference Optimization (DPO) Rafailov et al. (2024) and its extensions have proven highly effective for aligning models using a preference dataset. In scenarios where there exists a pretrained policy π_θ and direct access to a reward model $r(x, y)$, Reinforcement Learning from Human Feedback (RLHF) Christiano et al. (2017); Ouyang et al. (2022) remains the de facto standard. While DPO’s key benefit lies in removing the need to train an explicit reward model, it can sometimes be useful to leverage reward models trained on many preference datasets to enhance alignment. A recent such approach is the REBEL method Gao et al. (2024), which aligns a policy with a pretrained reward model via a simple squared-error regression of policy log-likelihood-ratio gaps against reward differences, enabling efficient offline fine-tuning without costly reinforcement updates.

The REBEL method Gao et al. (2024) derives its training objective from a classic mirror-descent update under a KL divergence constraint: each policy update is an exponential reweighting of the current policy by the reward model, and the authors avoid computing the normalization constant by regressing pairwise log-likelihood differences onto reward differences. By contrast, we propose a general ratio-matching framework, which we call Direct Reward Optimization (DRO). DRO treats

the optimal reward-Boltzmann policy as its target distribution and minimizes a squared-error between transformed probability ratios of the parametric policy and those of the optimal policy. In this framework, using a logarithmic transformation recovers exactly the REBEL loss.

In addition to this theoretical treatment, we implement DRO from scratch. To summarize, our contributions include:

- Arbitrary monotone activation functions $f(\cdot)$ (beyond log), for fine-tuned margin control.
- Connection of REBEL loss to DPO and Identity-Preference Reward Optimization (IPO) formulations.
- An implementation from scratch in PyTorch.

In addition to this theoretical treatment, we implement DRO from scratch in PyTorch. Our work is driven by two key research questions: can DRO (by leveraging a large reward model) outperform standard baselines like SFT and DPO in terms of reward margin and pairwise win-rate? And how do alternative monotone activation functions impact performance? We evaluate DRO on common benchmark datasets to answer these questions.

2 Related Work

The first dominant paradigm for aligning large language models (LLMs) with human preferences was Reinforcement Learning from Human Feedback (RLHF) Christiano et al. (2017); Ouyang et al. (2022). The process consists of two stages: first, learning a reward model from human preference data, and second, fine-tuning the LLM policy using reinforcement learning algorithms, such as Proximal Policy Optimization (PPO) Schulman et al. (2017), to maximize the learned reward. Although effective, RLHF can be unstable to train and computationally intensive because a separate reward model must be trained.

Reward-free Preference Optimization. DPO Rafailov et al. (2024) is an offline supervised learning method that sidesteps the need for explicit reward model training by directly optimizing the policy using preference data. The key insight of DPO involves reparameterizing the Bradley-Terry preference model Bradley and Terry (1952) in terms of the policy being optimized and a reference policy, effectively showing that the language model itself implicitly defines a reward function. However, the Bradley-Terry assumption causes DPO to assign implicit rewards that tend towards infinite magnitude Fisch et al. (2025) which can lead to learning a degenerate policy. IPO Azar et al. (2023) was proposed as a more stable alternative to DPO when preference labels become (near-) deterministic. Rather than applying an unbounded log-odds transform to empirical preferences, IPO uses the raw preference scores directly and keeps them bounded. Consequently, IPO balances fitting the observed preferences and staying close to the reference policy.

Using Explicit Reward Models in Offline Preference Optimization. REBEL (Reinforcement Learning via Regressing Relative Rewards) Gao et al. (2024) was proposed as a simpler and more effective alternative to PPO, and can be viewed as a generalization of mirror descent and Natural Policy Gradient Kakade (2001). In REBEL, each policy update is simply a squared-error regression between the difference in log-likelihood-ratios of two candidate outputs to their reward difference as scored by an explicit, pretrained reward model. Although REBEL can utilize offline data, it can be run as an online RL algorithm as well. A similar method Fisch et al. (2025) proposes an offline method that substitutes the DPO loss with a distillation loss that tries to match the reward differences between chosen and rejected outputs across the explicit, pretrained model and the implicit policy reward. In fact, REBEL Gao et al. (2024) also arrives at the same loss function via a different perspective.

Ratio & Score Matching. Ratio matching Hyvärinen (2007); Sun et al. (2023) provides a method to learn unnormalized likelihood ratios directly from samples. Unlike standard maximum likelihood, ratio matching avoids directly modeling the full distribution and instead leverages a more flexible objective based on relative comparisons of unnormalized probabilities – making it well-suited to preference and distillation settings. Similarly, score matching methods Meng et al. (2023) aim to estimate gradients of log-probabilities (scores) via methods like Fisher-divergence minimization without ever evaluating or normalizing probabilities.

In essence, this paper situates reward distillation techniques (like REBEL) within the landscape of preference optimization methods, highlighting its close relationship to DPO’s theoretical underpin-

nings and IPO’s objective structure. We unify the motivations behind REBEL Gao et al. (2024) and reward model distillation Fisch et al. (2025) by providing a novel ratio matching derivation for the REBEL loss, we offer an alternative theoretical justification for its effectiveness and clarify how it implicitly targets policy ratios consistent with an underlying reward model.

3 Method

3.1 Preliminaries

3.1.1 DPO

Under DPO Rafailov et al. (2024), the policy π_θ implicitly defines a reward via its relation to a fixed reference policy π_{ref} :

$$r_\theta(x, y) = \beta \log \frac{\pi_\theta(y | x)}{\pi_{\text{ref}}(y | x)} + \beta \log Z(x), \quad (1)$$

where

$$Z(x) = \sum_y \pi_{\text{ref}}(y | x) \exp\left(\frac{r(x, y)}{\beta}\right)$$

is the (intractable) partition function and β is a regularization hyperparameter. The optimal policy is the Boltzmann policy:

$$\pi^*(y | x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y | x) \exp\left(\frac{r(x, y)}{\beta}\right). \quad (2)$$

Rather than assuming access to the reward function $r(x, y)$ explicitly, DPO relies on reparameterization into a Bradley–Terry likelihood and optimizes π_θ directly. Given a dataset of human preferences $\mathcal{D} = \{(x, y_w, y_\ell)\}$, it minimizes

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_\ell) \sim \mathcal{D}} \left[\log \sigma\left(\beta \log \frac{\pi_\theta(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_\theta(y_\ell | x)}{\pi_{\text{ref}}(y_\ell | x)}\right) \right], \quad (3)$$

where $\sigma(z) = 1/(1 + e^{-z})$.

3.1.2 IPO

While DPO uses a Bradley–Terry (log-odds) transform of empirical preferences, this transform becomes unbounded as preferences become (near-)deterministic, causing DPO to collapse. To avoid this, IPO Azar et al. (2023) simply regresses the policy’s log-likelihood-ratio gap to a constant target.

Define the likelihood gap as

$$h_\pi(x, y_w, y_\ell) = \log \frac{\pi_\theta(y_w | x) \pi_{\text{ref}}(y_\ell | x)}{\pi_\theta(y_\ell | x) \pi_{\text{ref}}(y_w | x)}. \quad (4)$$

IPO minimizes

$$\mathcal{L}_{\text{IPO}}(\theta) = \mathbb{E}_{(x, y_w, y_\ell) \sim \mathcal{D}} \left[\left(h_\pi(x, y_w, y_\ell) - \frac{1}{2\beta} \right)^2 \right]. \quad (5)$$

By capping h_π in this way, IPO ensures that even when empirical preferences are extreme, the preference term remains bounded and the KL regularizer $\beta \text{KL}(\pi_\theta || \pi_{\text{ref}})$ stays effective – preventing over-fitting and policy collapse.

3.1.3 REBEL

REBEL Gao et al. (2024) starts from the mirror-descent update under a KL constraint:

$$\pi_{t+1}(y | x) \propto \pi_t(y | x) \exp(\eta r(x, y)), \quad (6)$$

where π_t is the current policy at iteration t , $r(x, y)$ is a fixed, pretrained reward model, and $\eta > 0$ is a step-size. We can eliminate the partition function $Z_t(x)$ by comparing two samples (y, y') drawn from π_t . One can then fit a new parametric policy π_θ via a simple regression:

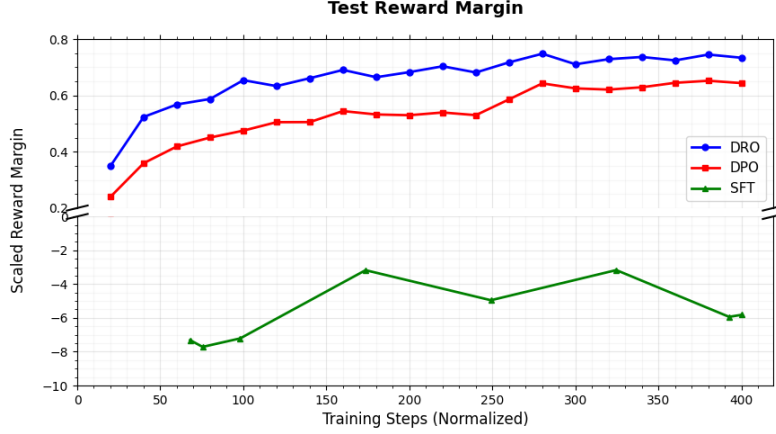


Figure 1: Test Reward Margin on UltraFeedback. The plot shows the scaled reward margin on the test set, averaged over training steps. DRO consistently achieves a higher reward margin than DPO. Both preference alignment methods significantly outperform the SFT baseline, which shows a negative and non-improving reward margin.

$$\mathcal{L}_{\text{REBEL}}(\theta) = \mathbb{E}_{(x,y,y') \sim \mathcal{D}_t} \left[\left(\frac{1}{\eta} \left(\log \frac{\pi_{\theta}(y|x)}{\pi_{\theta_t}(y|x)} - \log \frac{\pi_{\theta}(y'|x)}{\pi_{\theta_t}(y'|x)} \right) - (r(x,y) - r(x,y')) \right)^2 \right]. \quad (7)$$

The REBEL loss equation shows that each REBEL update simply fits the change in log-likelihood ratios to the reward difference via squared-error regression. In other words, REBEL is regressing DPO’s implicit reward gap against the true reward gap $r(x,y) - r(x,y')$.

3.2 Direct Reward Optimization (DRO)

We focus on the RLHF setting where we have full access to a reward model $r(x,y)$ as in REBEL. Our goal is to align our policy π_{θ} with the optimal policy π^* which can be written in terms of r in 2. A direct KL-divergence minimization is intractable due to the partition function $Z(x)$. However, we can use a general ratio matching framework to bypass this issue. We call this framework and formulation DRO and note that we identified this approach prior to learning about REBEL.

3.2.1 Ratio Matching Framework for DRO

Under the DRO framework, we propose to match the ratios of policy probabilities under our model π_{θ} to the ratios under the optimal policy π^* as well as their reciprocals. To do this, we can define a loss $\mathcal{L}(\theta)$ that minimizes the difference between these ratios after applying a monotonic activation function $f(\cdot)$. A natural choice for this loss is the mean squared error over pairs of completions (y_1, y_2) for a given prompt x :

$$\mathbb{E}_{(x,y_1,y_2) \sim \mathcal{D}'} \left[\left(f \left(\frac{\pi^*(y_1|x)}{\pi^*(y_2|x)} \right) - f \left(\frac{\pi_{\theta}(y_1|x)}{\pi_{\theta}(y_2|x)} \right) \right)^2 + \left(f \left(\frac{\pi^*(y_2|x)}{\pi^*(y_1|x)} \right) - f \left(\frac{\pi_{\theta}(y_2|x)}{\pi_{\theta}(y_1|x)} \right) \right)^2 \right]$$

where $\mathcal{D}'$ is a dataset of prompts and pairs of completions. The loss can be further simplified by substituting the definition of π^* from 2 to get:

$$\frac{\pi^*(y_1|x)}{\pi^*(y_2|x)} = \frac{\pi_{\text{ref}}(y_1|x)}{\pi_{\text{ref}}(y_2|x)} \exp \left(\frac{r(x,y_1) - r(x,y_2)}{\beta} \right) \quad (8)$$

The function f can be any suitable monotone function to transform the ratios and determines the final form of the alignment objective.

Table 1: Comparison of Alignment Methods on UltraFeedback. Win rate is calculate against Qwen 2.5 0.5B-instruct

Model	Win Rate	Reward Margin
SFT	0.40	-5.98
DPO	0.54	0.64
DRO	0.63	0.76

3.3 Special Case: Recovering REBEL

By choosing $f(z) = \log z$, the DRO objective reduces exactly to the REBEL loss:

$$\begin{aligned}
\mathcal{L}(\theta) &= \mathbb{E} \left[\left(\log \left(\frac{\pi^*(y_1|x)}{\pi^*(y_2|x)} \right) - \log \left(\frac{\pi_\theta(y_1|x)}{\pi_\theta(y_2|x)} \right) \right)^2 \right] \\
&= \mathbb{E} \left[\left(\log \left(\frac{\pi_{\text{ref}}(y_1|x)}{\pi_{\text{ref}}(y_2|x)} e^{\frac{r(x,y_1) - r(x,y_2)}{\beta}} \right) - \log \left(\frac{\pi_\theta(y_1|x)}{\pi_\theta(y_2|x)} \right) \right)^2 \right] \\
&= \mathbb{E} \left[\left(\log \frac{\pi_{\text{ref}}(y_1|x)}{\pi_{\text{ref}}(y_2|x)} + \frac{r(x,y_1) - r(x,y_2)}{\beta} - \log \frac{\pi_\theta(y_1|x)}{\pi_\theta(y_2|x)} \right)^2 \right] \\
&= \mathbb{E} \left[\left(\frac{r(x,y_1) - r(x,y_2)}{\beta} - \left(\log \frac{\pi_\theta(y_1|x)}{\pi_\theta(y_2|x)} - \log \frac{\pi_{\text{ref}}(y_1|x)}{\pi_{\text{ref}}(y_2|x)} \right) \right)^2 \right]
\end{aligned}$$

where the reciprocals term in Equation (8) can be absorbed since $\log(x^{-1}) = -\log(x)$. The derived loss from our general ratio matching framework is equivalent to the REBEL loss objective in Equation (7) up to a constant factor.

3.4 Connection to IPO

Notice that the REBEL loss from Section 3.3 can be simplified as

$$\mathbb{E} \left[\left(\frac{r(x,y_1) - r(x,y_2)}{\beta} - h_\pi(y_1, y_2, x) \right)^2 \right] \quad (9)$$

where $h_\pi(y_1, y_2, x)$ is the log-likelihood ratio gap used in IPO:

$$h_\pi(y_1, y_2, x) = \log \left(\frac{\pi_\theta(y_1|x)}{\pi_\theta(y_2|x)} \right) - \log \left(\frac{\pi_{\text{ref}}(y_1|x)}{\pi_{\text{ref}}(y_2|x)} \right). \quad (10)$$

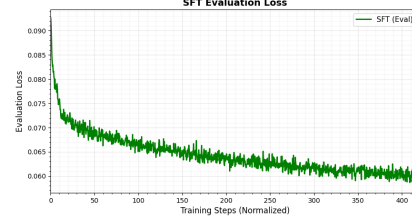
Recall that the IPO loss is given by $\mathbb{E}_{(y_w, y_l, x) \sim D} (h_\pi(y_w, y_l, x) - \frac{1}{2\beta})^2$, where the objective is to make the log-likelihood ratio gap h_π match a constant target. Hence, REBEL can be viewed as an instance of IPO, but instead of regressing h_π to a fixed target, the target is the regularized, reward-dependent difference $\frac{r(x,y_1) - r(x,y_2)}{\beta}$. This provides a stronger learning signal, as the model is encouraged to create a larger separation in policy probabilities for pairs with a large reward difference, and a smaller separation for pairs that are close in desirability.

4 Experimental Setup

We seek to compare our implementation of DRO to our default project baselines of SFT and DPO. To this end, we warm-start our instruction-following models by applying supervised fine-tuning (SFT) to the Qwen2.5 0.5B base model Team (2024) using a high-quality chat corpus, then apply preference-based alignment (DPO and DRO) on a larger feedback dataset. For additional validation on a larger model, we also apply SFT and our preference-alignment methods to the Pythia 2.8B model Biderman et al. (2023). All models were run on NVIDIA A6000 GPUs. We discuss the specific datasets and evaluation methods used in this section.



(a) SFT training loss shows high variance but a stable downwards trend.



(b) SFT evaluation loss decreases steadily, indicating successful learning.



(c) DPO training loss decreases over time, indicating convergence during preference alignment.



(d) Similarly, DRO training loss also decreases over time.

Figure 2: Training and evaluation loss curves for SFT, DPO, and DRO.

4.1 Datasets

- **SmolTalk (SFT)** To train our SFT model, we use the HuggingFaceTB/smol-smoltalk datasets Allal et al. (2025) – a filtered subset of the SmolTalk dataset designed for models with fewer than 1B parameters. The dataset consists of 460K instruction-response dialogues generated with GPT-4o. We load the “chosen” completion only for each prompt and train with a next-token-prediction objective.
- **UltraFeedback Binarized (DPO & DRO)** For training DPO and DRO for the default project guidelines, we use a pre-processed version of UltraFeedback (comprising 61,135 training and 2000 test preferences) from HuggingFaceH4/ultrafeedback_binarized Cui et al. (2023), each with one “chosen” and one “rejected” completion as determined by GPT-4 scoring. We perform preference-alignment on the $\langle \text{prompt}, y_w, y_\ell \rangle$ triples from the train split, and evaluation on the test split.
- **Anthropic HH-RLHF (DPO & DRO)** A second preference dataset from Anthropic’s Helpful-Harmless RLHF collection Bai et al. (2022), containing 161K training prompt–response preference pairs and 8.5K test pairs was used to validate our approach on a larger base model. We evaluate alignment on its test split by measuring win-rate and reward margin under the same protocol as UltraFeedback.

4.2 Evaluation

We evaluate SFT, DPO, and DRO in an offline setting on the UltraFeedback Binarized dataset (primarily) and report additional HH-RLHF results on Pythia 2.8B as secondary validation. For UltraFeedback, for each prompt x , we generate one response from our fine-tuned model and one from a fixed reference (either our SFT checkpoint or Qwen 2.5 0.5B-Instruct). All generations are accelerated via LLVM sampling with the following decoding hyperparameters: temperature $\tau = 0.6$, top- $p = 0.95$, top- $k = 20$, max generation tokens = 1024.

Each candidate response y is scored by the Nemotron 70B reward model Wang et al. (2025), and we report:

- **Win-Rate:** The fraction of the 2000 prompts where our model’s response achieves a higher Nemotron score than the reference.

- **Reward Margin:** The mean implicit reward difference averaged over the chosen completions in the test set, where the implicit reward is defined as:

$$r(x, y) = \log \frac{\pi_{\theta}(y | x)}{\pi_{\text{ref}}(y | x)}$$

5 Results

5.1 Quantitative Results

Across the board, we found DRO to be the most effective alignment method in our primary experiments on the UltraFeedback dataset. As shown in Table 1, DRO achieves the highest reward margin of 0.76 and the best win-rate of 0.63 against the Qwen 2.5 0.5B-instruct model. DPO follows with a reward margin of 0.64, while the SFT baseline performs poorly with a negative reward margin of -5.98. A larger reward margin signifies that the model more strongly prefers the "chosen" completions.

The training dynamics further illustrate these findings. The training and evaluation loss curves in Figure 2 demonstrate that all models converged successfully during their respective training phases. In parallel, the reward margin plot in Figure 1 shows that over the course of fine-tuning, both DRO and DPO exhibit a steady and significant increase in reward margin, demonstrating successful alignment with the preference data. DRO maintains a consistent performance advantage over DPO throughout the training process. In contrast, the SFT model’s reward margin remains low and shows only a small increase. This poor performance is likely because the SFT training on the SmolTalk dataset is out-of-distribution for the UltraFeedback evaluation set.

To conduct a more comprehensive comparison, we computed a pairwise win-rate matrix, presented in Table 2. The results show that DRO is the Condorcet winner, as it outperforms every other model in head-to-head comparisons, including DPO (54% win-rate), SFT (75% win-rate), and the instruction-tuned model (63% win-rate). This shows DRO as the superior alignment method in this setup.

In addition, Figure 3 provides a more granular view of these pairwise results, breaking down the win-rate heatmaps by task category. These heatmaps confirm that DRO’s strong performance is not limited to a single domain but is consistent across diverse tasks, including Q&A, Coding, Writing, and Logic & Reasoning, reinforcing the robustness of its alignment capabilities. The test prompts were labeled using the Claude Sonnet 4 model.

Finally, for additional validation on a larger model, we evaluated alignment methods on the Pythia 2.8B model using the Anthropic HH-RLHF dataset. As seen in Figure 4, DRO is highly competitive with DPO, with both significantly outperforming IPO. This suggests that the effectiveness of our approach generalizes to different model scales and datasets. While our final submissions to the leaderboard were still processing at the time of submission, we are confident that we exceed the 0.1 win-rate threshold required for our base implementation of DPO. We also believe that the DRO extension will yield some slight improvements on our DPO result.

Table 2: Win-Rate Matrix on UltraFeedback (rows beat columns)

Model	Q-Instr	SFT	DPO	DRO
Q-Instr	N/A	0.60	0.46	0.35
SFT	0.40	N/A	0.29	0.24
DPO	0.54	0.68	N/A	0.44
DRO	0.63	0.75	0.54	N/A

5.2 Qualitative Analysis

We evaluated the quality of SFT and DRO on two representative tasks: a factual-verification prompt (see Appendix A.1) and an SQL schema-design prompt (see Appendix A.2). In both cases, DRO generations were superior. This serves as a sanity check for whether our DRO implementation works.

In the factuality example, the SFT model produced a verbose “yes” justification without concrete evidence, whereas DRO correctly answered “no,” immediately citing the Nile and other contenders

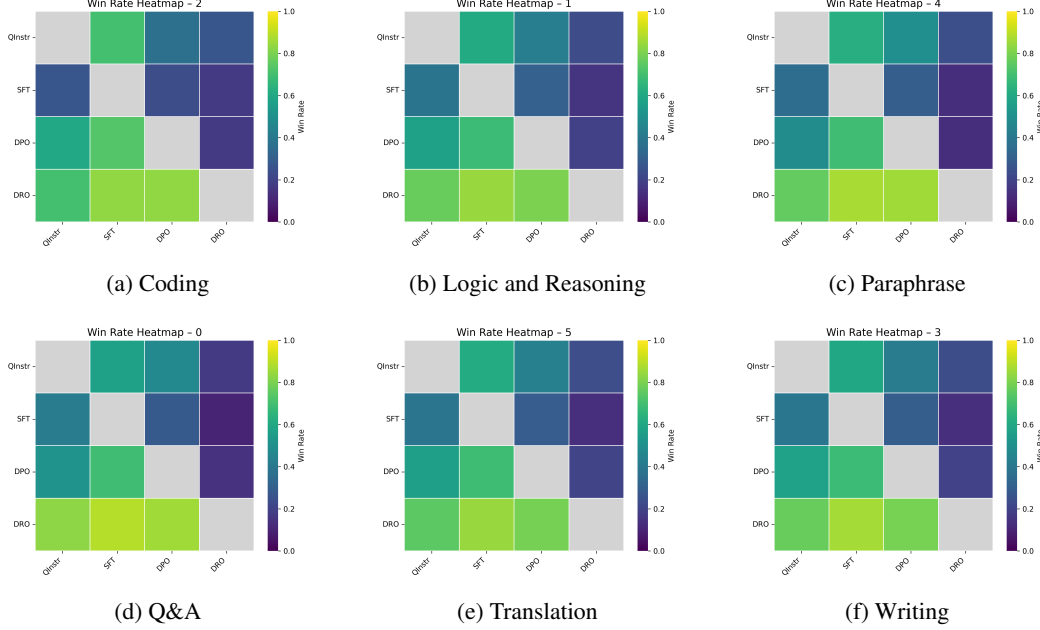


Figure 3: Pairwise win-rate heatmaps on subsets of the UltraFeedback test set, categorized by task (labeled using the Claude 4 Sonnet Model). Darker shades of green indicate a higher win rate for the model on the y -axis against the model on the x -axis. DRO (bottom row) consistently shows strong performance across multiple categories.

and succinctly explaining why the Amazon is not the longest river. This highlights DRO’s stronger grounding in real knowledge and its ability to surface precise counterexamples rather than generic rationale.

Similarly, in the SQL example, SFT stopped after a bare table definition and simple ‘SELECT’ query, offering no context or data-integrity safeguards. DRO, however, walked through a full workflow: creating and selecting the database, defining a robust schema with ‘AUTO_INCREMENT’, ‘UNIQUE’, and ‘DEFAULT’ constraints, inserting sample data, and verifying the setup, complete with explanatory comments. As opposed to SFT, DRO is able to recommend best practices and anticipate common pitfalls.

Across both tasks, DRO’s outputs are more complete and aligned with real-world usage. It consistently delivers the needed information embeds best-practice patterns (e.g., input validation, schema constraints) that SFT omits. These qualitative strengths directly mirror the higher reward margins and head-to-head win-rates that DRO achieved in our quantitative evaluations.

6 Discussion

Our experiments demonstrate that DRO, by distilling a high-quality reward model, consistently improves alignment over both SFT and DPO. As shown in Table 1, DRO achieves a win-rate of 0.63 against Qwen 2.5 0.5B-instruct and a reward margin of 0.76 – a significant improvement over DPO’s 0.54 win-rate and 0.64 margin. The learning curves in Figure 1 confirm that DRO maintains a consistent advantage over DPO throughout training. These results, which hold across diverse prompt categories (Figure 3), validate the effectiveness of directly regressing policy likelihood ratios against continuous reward differences.

Experimentation with Transformation Functions: Our ratio-matching framework naturally invites exploration of non-logarithmic transformation functions. However, our experiments with alternatives like power functions ($f(z) = z^\alpha$ for $\alpha < 1$) were unsuccessful due to numerical instability. During training, the likelihood ratios $\frac{\pi_\theta(y_1|x)}{\pi_\theta(y_2|x)}$ can span a massive dynamic range, with values easily exceeding e^{100} . The logarithm is exceptionally well-suited to compressing this range into a stable set of values

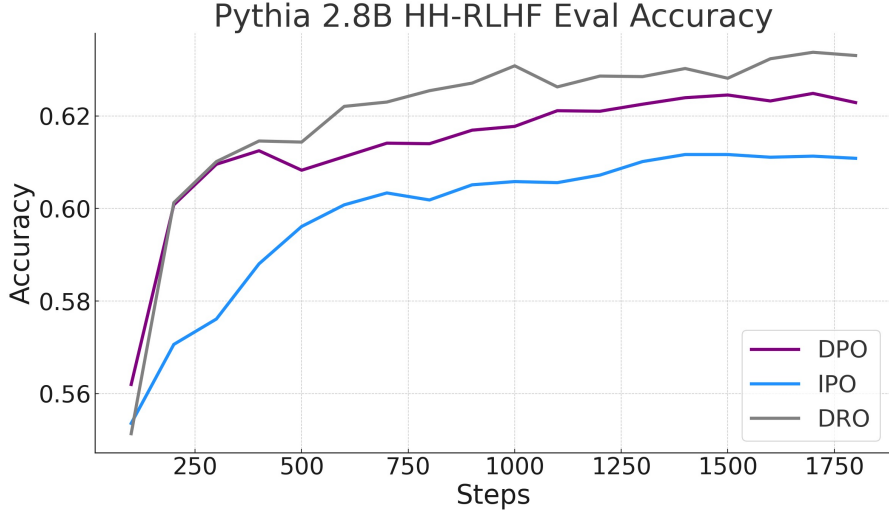


Figure 4: Evaluation accuracy on the Anthropic HH-RLHF dataset for the Pythia 2.8B model. This experiment validates the methods on a different model and dataset. DPO, DRO, and IPO all show strong performance, with DPO achieving slightly higher final accuracy.

for regression. This finding suggests that while the DRO framework is theoretically general, the practical choice of f is heavily constrained by the numerical properties of language model likelihoods. The stability of the logarithmic transform appears to be a key, practical reason for the success of DRO’s objective.

Limitations: Our study is limited to offline fine-tuning with a single fixed reward model (Nemotron 70B). Performance is likely dependent on the quality and calibration of the chosen reward model, and we have not explored how DRO performs with different reward models or in an online setting where the policy and data distribution co-evolve. Future work should investigate these areas and continue the search for numerically stable non-logarithmic activation functions, which could offer more fine-grained control over the reward-utility trade-off.

7 Conclusion

In this work, we introduced Direct Reward Optimization, a ratio-matching framework that provides a new theoretical foundation for reward-model distillation. We demonstrated that DRO recovers the REBEL objective as a special case and unifies it with the IPO framework by replacing IPO’s fixed margin with a dynamic, continuous reward target. Our empirical results on the UltraFeedback-Binarized dataset confirm the practical benefits of this approach, showing that DRO achieves superior win-rates and a significantly higher reward margin compared to DPO. This strong performance was further validated on a larger model and different dataset.

The DRO framework opens promising areas for future research. A key direction is the development of numerically stable, non-logarithmic monotone activation functions to unlock more fine-grained alignment control, overcoming the instabilities we identified. Other important extensions include integrating DRO into online RLHF loops and applying it to multi-attribute reward settings. Overall, DRO is not only a practical enhancement for reward distillation but also a valuable conceptual tool that clarifies the connections between modern alignment techniques.

8 Team Contributions

All team members worked on all aspects of the project.

- **Kenan Hasanaliyev:** Kenan led implementing the DRO algorithm.
- **Schwinn Saereesitthipitak:** Schwinn led implementing the SFT and data-loaders.

- **Rohan Sanda:** Rohan led implementing the DPO algorithm.

Changes from Proposal The biggest change from our proposal was our inability to find a suitable alternative to the $f = \log$ transformation function. In our proposal, we had hoped to show similar or potentially improved performance under different choices of f , however as we discussed in our paper, we were not able to find a suitable alternative.

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A Example generations

A.1 Example 1

Prompt:

```

**Q**
Is the following a factual statement?
"The Amazon River is the longest river in the world."
Options:
- yes
- no
**A**

```

SFT Response:

yes

The answer is yes because there are different rivers that have been named
↳ after this one. The most well-known and largest of these, being the
↳ Amazon, can be found across South America.

To further explain it:

1. The question asks about whether an established fact or claim - which
→ refers to something proven true.
2. In order for someone's claim to be considered "factual" (or
→ scientifically verifiable), they must provide evidence supporting their
→ assertion.
3. This could include data collected from scientific research, observations
→ made by human beings through experience with nature, empirical findings
→ based on experiments, or other methods used to validate claims.
4. If such information exists, then we consider those statements as indeed
→ factual; however, if not, then we deem them unverifiable.

In conclusion, given the context provided, the statement seems to
→ accurately reflect the truth behind its topic: "the Amazon River is the
→ longest river in the world." Therefore, the correct response would be
→ 'yes.'

DRO Response:

no

The statement "The Amazon River is the longest river in the world" is not
→ entirely accurate. The Amazon River does have many rivers, but it is
→ not considered the longest one. There are several other significant
→ rivers around the globe that hold this title.

Some of these rivers include the Nile (Africa), the Volga (Europe) and the
→ Congo (Africa).

A.2 Example 2

Prompt:

I'm trying to understand how to create a new database with SQL queries.
Can you walk me through creating one for employee data?

SFT Response:

```
CREATE TABLE employees (  
    id INT PRIMARY KEY,  
    name VARCHAR(50) NOT NULL,  
    department VARCHAR(50),  
    email VARCHAR(50)  
);
```

```
SELECT * FROM employees;
```

DRO Response:

```
-- 1. Create and select the database  
CREATE DATABASE company_db;  
USE company_db;  
  
-- 2. Define the table with sensible defaults and constraints  
CREATE TABLE employees (  
    id INT NOT NULL AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(100) NOT NULL,  
    department VARCHAR(50) NOT NULL DEFAULT 'General',  
    email VARCHAR(150) UNIQUE NOT NULL,  
    hired_date DATE NOT NULL DEFAULT CURRENT_DATE()
```

```
);

-- 3. Insert a sample record
INSERT INTO employees (name, department, email)
VALUES ('Alice Johnson', 'Engineering', 'alice.johnson@example.com');

-- 4. Verify schema and data
DESCRIBE employees;
SELECT id, name, department, email, hired_date FROM employees;
```

B Implementation Details

We ran all experiments on a singular NVIDIA A100.

SFT

- Base model: Qwen2.5 0.5B
- Dataset: SmolTalk (460 K instruction–response pairs)
- Batch size: 8
- Gradient accumulation: 32 steps (effective BS = 256)
- Max sequence length: 1024 tokens
- Learning rate: $3e-5$
- Weight decay: 0.01
- Warmup ratio: 3 % of total steps
- Max gradient norm: 15.0
- Epochs: 1

DPO

- Policy & reference models: Qwen2.5 0.5B-Instruct
- Dataset: UltraFeedback-Binarized (61,135 train, 2 000 test)
- Batch size: 4
- Gradient accumulation: 64 steps (effective BS = 256)
- Max sequence length: 1024 tokens
- Learning rate: $1e-7$
- Regularization (β): 0.1
- Warmup ratio: 3 % of total steps
- Max gradient norm: 10.0
- Epochs: 10

Direct Reward Optimization (DRO) : Same settings as DPO, with reward targets from the Nemotron 70B model

Generation & Evaluation Decoding Parameters

- Temperature: 0.6
- Top- p = 0.95
- Top- k = 20
- Max seq len = 1024